Measuring and modeling perceived humanness

MDI Scholar: Rebecca Ansell, M.S. Computer Science, May 2025
Research Team: Lisa Singh, Leticia Bode, Tiago Ventura, Sejin Paik, Autumn Toney-Wails

Research Goals:

Our project aims to understand what drives our perception of humanness of text, how we can measure it and if we can model it.

Our Studies:

- **Study 1 (Fall, 2024):** Initial analysis of how we perceive, measure and model perceived humanness on X and YouTube
- **Study 2 (Spring, 2025):** *Human-in-the-Loop* - investigating methods for fine-tuning for improved perceived humanness
- **Current Work 1:** Extended analysis to four social media platforms
- **Current Work 2:** Fine-Tuning LLMs to improve their perceived humanness by leveraging reinforcement learning and our prediction model

Methodology

1. Data Collection: social media posts discussing the 2024 Presidential Debate
   - Real World Posts Data
     - Collected from X, YouTube, TruthSocial and TikTok through debate related keywords
   - Generated Posts Data
     - Generated posts from Llama and GPT models

2. Preprocessing:
   - Data cleaning:
     - General formatting, censoring usernames and hyperlinks

3. Data Labelling:
   - Annotators were given pairwise comparisons of different post combinations and labelled which sounded more like AI

4. Ranking Posts:
   - Using Bradley-Terry we ranked posts from most AI-like (-1) to most human-like (+1) from human preferences
   - Results in a full dataset of ranked perceived humanness posts, as well as a dataset of preference pairs, ready for analysis and modeling.

5. Developing Generating Model
   - Using ranked posts to fine-tune LLMs to generate text that is perceived as more humanlike

Example Labeling Task:

Which of the following two X Posts is more likely to be generated by an AI Chatbot?

1. why do politicians always avoid direct answers?? 😒 #Debatenight" - X Post
2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr - ChatGPT 4o

Perceived Humanness Comparison

- LLaMA 3.3 bridges the humanness gap, often matching or exceeding human-authored margins across platforms.
- TikTok exhibits the largest distinction in perceived humanness between human posts and LLM-generated content.
- Offensive and negative language correlates with human authorship. Positive tones are more nuanced with optimistic tone tending to be attributed to AI-generated text and joy to human text.

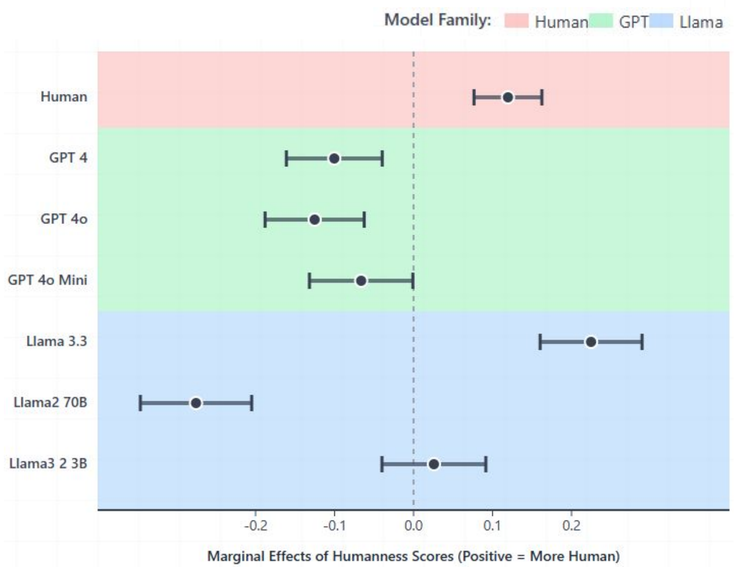


Figure 1: Perceived humanness scores across platforms and authors. This study was conducted on the Connect Platform, with 1,012 participants across two surveys. The demographics of the participants were representative of the broader U.S. population.

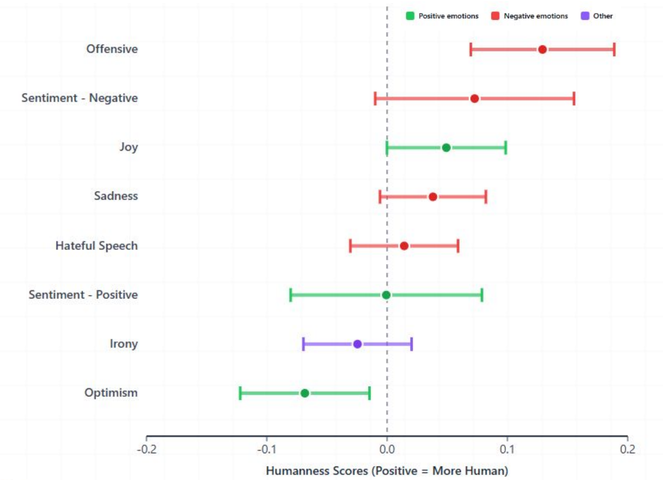


Figure 2: TweetEval affect analysis showing tone shifts across perceived humanness ranks

PHI-Score

- Perceived Humanness Indicator (PHI) is a continuous score between −1 (most AI-like) and +1 (most human-like) derived from our Bradley-Terry method.
- We can use this score to fine tune for perceived humanness.

Post	PHI Score
#Debate2024 goal should be to dismantle the establishment—let's keep fighting for an America-first strategy!	-0.854
Dems on economy = JOBLESSNESS, INFLATION, MISERY. We need TRUMP BACK! #Debatenight	0.837

Prediction Results

- PHI-Predict: We trained separate regression models using the posts and their PHI-Scores
- We excluded our lowest confidence posts to decrease noise and subjectivity ($-0.25 < \Phi < 0.25$)

Platform	Mean Squared Error
X	0.175
YouTube	0.158
TruthSocial	0.165
TikTok	0.187

Limitations

- Subjective judgments: Annotators often disagreed on what felt human, and platform context (X vs. YouTube) influenced perception.
- Our prediction model is trained on a very small sample size with limited resources because of only using high confidence posts.

Ethical Considerations

- We censored all usernames and hyperlinks for the labeling surveys to protect user privacy.
- We purposefully generated misinformation to emulate real social media posts for our research study only.

Future Work

- Using our PHI-Predict as our reward model, we would like to train a reinforcement learning model to improve the perceived humanness of text
- Extending our analysis on how humans perceive humanness

Acknowledgements

Thank you to the MDI Scholars program for an incredible opportunity to engage in meaningful research as a master's student. Thank you to Lisa, Leticia, Tiago, Sejin, Becky, Autumn and Kenny for your continued collaboration, and support. Thank you to MDI, the McCourt School and the Tech and Public Policy Grants Program for supporting this work.

Measuring and modeling perceived humanness

MDI Scholar: Rebecca Ansell, M.S. Computer Science, May 2025
Research Team: Lisa Singh, Leticia Bode, Tiago Ventura, Sejin Paik, Autumn Toney-Wails

Perceived humanness is the subjective impression that a piece of content reflects human-like authorship and captures how individuals interpret the linguistic properties that define humanness of written human communication.

Abstract

AI-generated text is becoming more prevalent in online discourse. Our research looks at how human annotators perceive humanness, and how we can model and measure it over various studies.

Our Studies:

- **Study 1 (Fall, 2024):** Bradley–Terry ranking on X vs. YouTube: initial analysis on how we perceive and measure humanness
- **Study 2 (Spring, 2025):** *Human-in-the-Loop* - investigating methods for fine-tuning for improved perceived humanness
- **Current Work 1:** Extended analysis to four social media platforms
- **Current Work 2:** Fine-Tuning LLMs to improve their perceived humanness by leveraging reinforcement learning and our prediction model

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Methodology

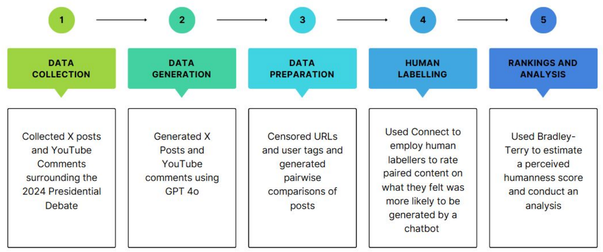


Figure 1: Data Collection Methodology

Example Labeling Task:

Which of the following two X Posts is more likely to be generated by an AI Chatbot?

1. "why do politicians always avoid direct answers?? 😒 #Debatenight" - X Post
2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr - ChatGPT 4o

Results (Analysis)

- LLaMA 3.3 bridges the humanness gap, often matching or exceeding human-authored margins across platforms.
- TikTok exhibits the largest distinction in perceived humanness between human posts and LLM-generated content.
- Offensive and negative language correlates with human authorship, whereas optimistic tone tends to be attributed to AI-generated text.

Figures

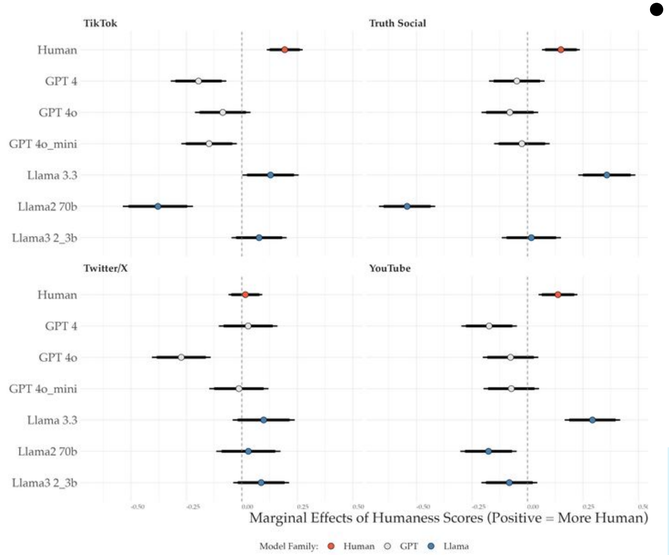


Figure 2: Perceived humanness across platforms and authors

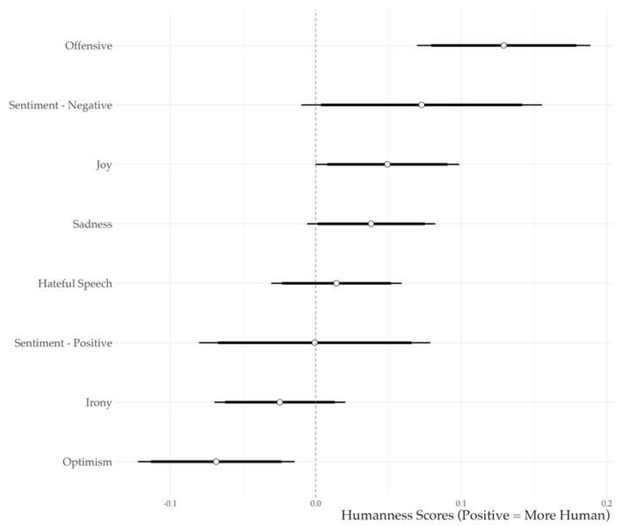


Figure 3: TweetEval affect analysis showing tone shifts across perceived humanness ranks

Results (Prediction)

- PHI-Predict: We trained separate regression models on our highest-confidence, ranked posts to predict a humanness score from –1 (AI-like) to +1 (human-like).

Platform	MSE
X	0.175
YouTube	0.158
TruthSocial	0.165
TikTok	0.187

Ethical Considerations

- We censored all usernames and hyperlinks for the labeling surveys to protect user privacy.
- We purposefully generated misinformation to emulate real social media posts for our research study only.

Limitations

- Subjective judgments: Annotators often disagreed on what felt human, and platform context (X vs. YouTube) influenced perception.
- Our prediction model is trained on a very small sample size with limited resources because of only using high confidence posts.

Future Work

- Using our PHI-Predict as our reward model, we would like to train a reinforcement learning model to improve the perceived humanness of text
- Extending our analysis on how humans perceive humanness

Detecting Humanness across Social Media Platforms

MDI Scholar: Rebecca Ansell, MS CS, 2025
Research Team: *The Humanness Project,* Dr. Lisa Singh, Dr. Leticia Bode, Dr. Tiago Ventura

Abstract

Humanness refers to the degree to which a piece of content is perceived to be generated by a human rather than by an artificial intelligence (AI). This research investigates how well humans can distinguish AI generated posts, from human authors, and the effectiveness of various embeddings and neural network architectures for classifying humanness using social media data from YouTube and X. We evaluated multiple models with different embeddings. The performance was assessed using relevant metrics. Our findings highlight the potential of specific embeddings and network configurations in predicting humanness with moderate accuracy, enabling insights into the interpretability of perceived humanness in social media posts.

Introduction

In the ever-evolving landscape of social media and generative artificial intelligence, distinguishing between human-written content and artificial intelligence (AI) generated text has become increasingly important. This study focuses on YouTube comments and X posts related to the 2024 election cycle.

Objectives:

- Analyze the perceived humanness of these posts using human annotators by using Bradley-Terry and ELO rankings to determine the most human-like posts.
- Building models to detect perceived humanness.

Research Questions:

1. Are humans able to detect AI-generated posts?
2. Are machine learning models able to detect AI-generated posts?
3. How well can we create models to detect perceived humanness?

This approach allows for an exploration of AI's influence on social media public's ability to discern human from non-human textual contributions.

Acknowledgements

Thank you to Dr. Lisa Singh, Dr. Leticia Bode, Dr. Tiago Ventura, and Rebecca Vanarsdall for your guidance, teaching, and ideas. Thank you to Kenny Ahmed for help with data collection, and to the REU students for data labelling. Thank you to the Massive Data Institute and the McCourt School for giving me the opportunity to work on a project like this. Thank you to the Tech and Public Policy Grants Program for supporting this work.

Methodology

1. Data Collection:
- Human Posts Data
  - Collected from X and YouTube through keywords
- Generated Posts Data
  - Generated from ChatGPT 4o and 3.5 using the same keywords

2. Data Labelling:
- Annotators were given pairwise comparisons of different post combinations and labelled which sounded more like AI
- Way to measure perceived humanness

3. Preprocessing:
- Data cleaning:
  - General formatting, censoring usernames and hyperlinks
- Ranking the posts
  - Using Bradley-Terry and ELO to rank posts from most human-like to most AI-like

4. Model Building
- Created multiple models with different embedding strategies

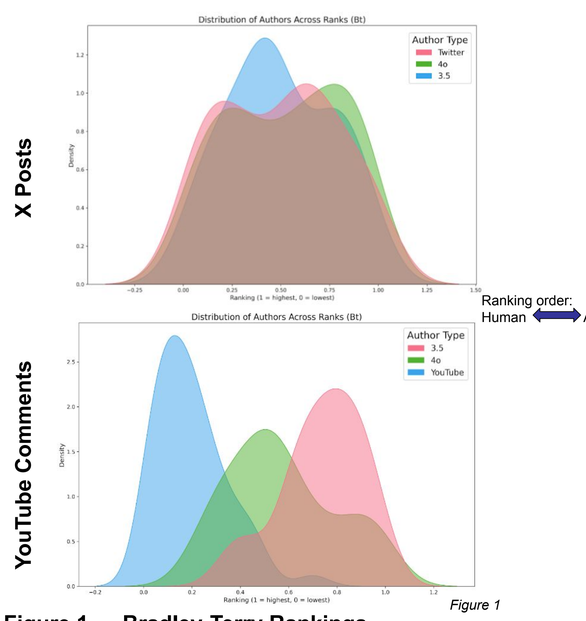


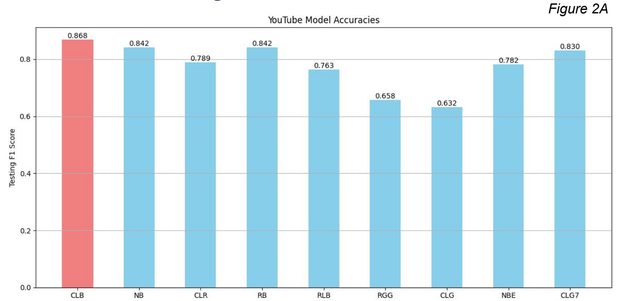
Figure 1

Figure 1 — Bradley-Terry Rankings
Using the BT Model, posts have been ranked based off of their pairwise comparisons results from most perceived as human to most perceived as AI. We can see that for X the posts are very difficult to distinguish, whereas YouTube appears to be well discerned by annotators. All pairs were double-labelled and there was a 49% disagreement between annotators.

Example of an X labelling task:
Which of the posts (Post A or Post B) is more likely to have been generated by an AI Model?
A: "Ensuring election integrity is crucial for democracy. #Democracy"
B: "I thought mail in voting was all fraudulent? If it's rigged, why do we vote?"

Preliminary Results

Figure 2A



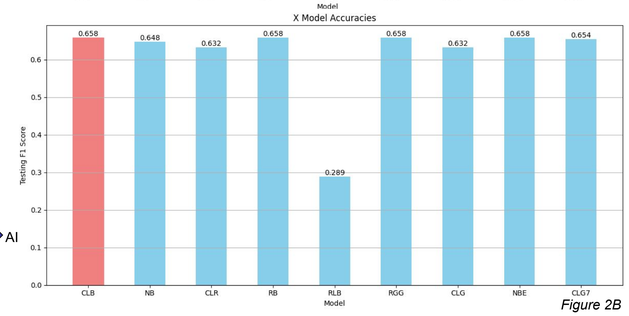


Figure 2B

Abbreviation	Model
CLB	CNN with LSTM (BERT)
NB	Perceptron NN (BERT)
CLR	CNN with LSTM (RoBERTa)
RB	RNN (BERT)
RLB	RNN with LSTM (BERT)
RGG	RNN with GRU layer (GPT-2)
CLG	CNN with LSTM (GPT-2)
NBE	Perceptron NN (BERT) (ELO ranking)
CLG7	CNN with LSTM (gemma-7b)

Figure 2 — Classifier Performance
2A YouTube: This figure shows the Testing F1 scores of various models applied to YouTube data.
Figure 2B: This figure shows the Testing F1 scores of various models applied to X data.

Discussion:

From the density graphs, We can see that for X the posts are very difficult to distinguish, whereas YouTube appears to be well discerned by annotators. The best model for both YouTube comments and X posts combines BERT embeddings with CNN for local pattern recognition and LSTM for sequential context. This model performed well because it effectively captures both local features and long-term dependencies in the text, leveraging the powerful contextual embeddings provided by BERT.

Limitations

- Limited labelled data covering election topics that are constantly changing.
- Human annotators had a difficult time differentiating between human and AI-generated content from X, this has limited our model building so far.
- We could not generate misinformation from any other chat bots except for GPT models, a greater diversity of chatbot sources would allow us to have a more diverse set of data.
- The election candidates have changed since the original data collection, we will need to expand our data collection and labeling to account for the change in candidates. We will also need to collect data on new topics that have emerged since the original data collection phase.

Ethical Considerations

- We censored all usernames and hyperlinks for the labeling surveys to protect user privacy.
- We purposefully generated misinformation to emulate real X posts and YouTube comments for our research study only.

Future Directions

1. We will use these models on election 2024 data to assess relative levels of humanness on different platforms.
2. Expand data collection to include the new democratic party's presidential nominee.
3. Expand study to other platforms: Reddit, TruthSocial.
4. Generate data from a wider variety of chatbots.
5. Further fine tuning and hyperparameter grid searching of current models.
6. Using more advanced embedding strategies such as Llama3.1, gemma-27b, and bloom-560m.

References

1. Carlson, David, and Jacob M. Montgomery. "A Pairwise Comparison Framework for Fast, Flexible, and Reliable Human Coding of Political Texts." *American Political Science Review*, vol. 111, no. 4, 2017, pp. 835-843, https://doi.org/10.1017/S0003055417000302.
2. Jia, Chenyan, and Taeyoung Lee. "Journalistic Interventions Matter: Understanding How Americans Perceive Fact-Checking Labels." *Harvard Kennedy School (HKS) Misinformation Review*, vol. 5, no. 2, 2024, https://doi.org/10.37016/mr-2020-138.
3. Routh, Jennifer, et al. "The Use of Pairwise Comparisons and the Elo Algorithm to Quantitatively Rate a Large Number of Items in a Single Dimension." EdArXiv, 16 Dec. 2022. Web.

Measuring and Modeling Perceived Humanness in AI-Generated Political Discourse

MDI Scholar: Rebecca Ansell, M.S. Computer Science, May 2025
Research Team: Lisa Singh, Leticia Bode, Tiago Ventura, Sejin Paik, Autumn Toney-Wails

Perceived Humanness: The extent to which a post exhibits natural, contextually appropriate, and stylistically nuanced characteristics associated with human writing.

Research Questions

1. Can people distinguish between human- and AI-generated social media posts?
2. What stylistic features drive perceptions of humanness?
3. Can we fine-tune chatbots to sound more human-like?

Motivation

Political social media content plays a major role in shaping opinion. AI language models are now part of this online discourse. But how much of the conversation is perceived to be human?

Example Labeling Task:

Which of the following two X Posts is more likely to be generated by an AI Chatbot?

1. "why do politicians always avoid direct answers?? 😒 #Debatenight" - X Post
2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr - ChatGPT 4o

Acknowledgements

Thank you to Lisa, Leticia, Tiago, Sejin, Becky, Autumn and Kenny for an amazing opportunity and year of collaboration, support, and growth. Thank you to MDI, the McCourt School and the Tech and Public Policy Grants Program for supporting this work.

Study 1: Assessing Perceived Humanness

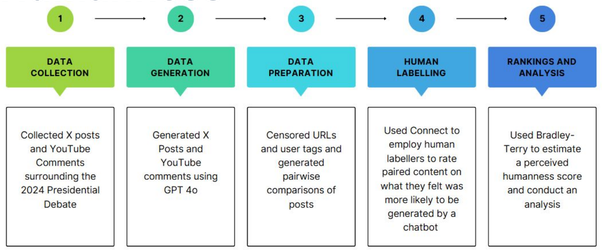


Figure 1: Data Collection Methodology

Ranking Perceived Humanness

- Labellers could generally distinguish between AI-generated and human-authored content
- Separation on X is more than on YouTube

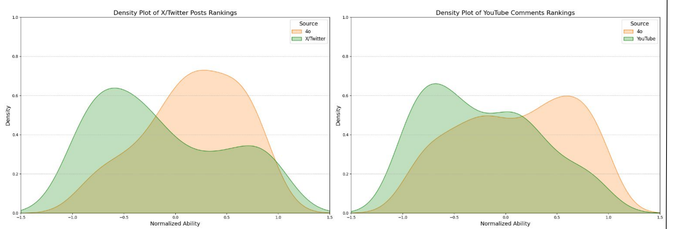


Figure 2: Density plot of PHI scores by model type, showing human posts often outscored fine-tuned models.

Affectability Results

- Positive sentiment often perceived as AI-authored.
- Negative sentiment and offensive language linked to human authorship.

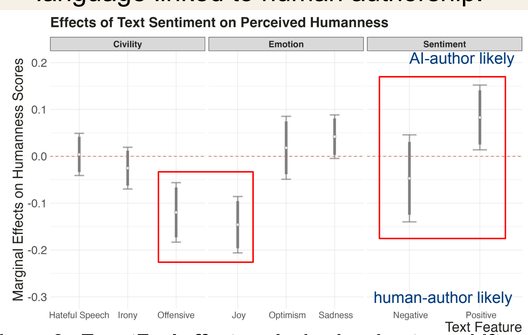


Figure 3: TweetEval affect analysis showing tone shifts across perceived humanness ranks

Study 2: Fine-Tuning with Perceived Humanness

Using data from Study 1 allowed us to model perceived humanness in 3 ways:

Method	Training Data			Used For
1	Prompt	Chosen	-	Supervised Fine Tuning
2	Prompt	Chosen	Rejected	Direct Preference Optimization Fine Tuning
3	Post	PHI-Score	-	PHI-Predict Regression Tool

Key Findings

➔ **More tuning ≠ more human:** Adding preference data often reduced humanness; DPO altered style but didn't improve perception.

➔ **Perception is subjective:** Annotator judgments varied, and text features alone weakly predicted humanness.

➔ **Smaller, cleaner data helps:** A DPO subset trained only on human-preferred pairs produced more natural outputs.



Figure 4: Human Evaluation of Fine-Tuned Models

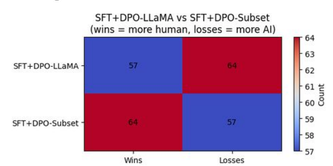


Figure 5: DPO-Subset Outperforms DPO in Human Judgments
The DPO-Subset model was trained only on comparisons where **human-authored posts were preferred over AI**, providing a clearer training signal.

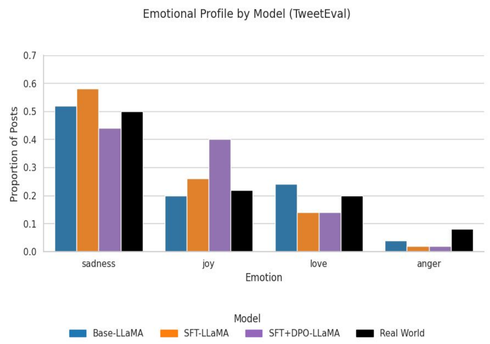


Figure 6: Emotional profile of model outputs based on TweetEval.
All models skewed toward sadness, with SFT+DPO generating the most joy. Anger was rarely expressed across all systems, suggesting limited emotional range in that category.

Limitations

- Subjective judgments: Annotators often disagreed on what felt human, and platform context (X vs. YouTube) influenced perception.
- Fine-tuning tradeoffs: More data didn't always improve humanness.
- Weak predictions: Text-based models couldn't reliably estimate perceived humanness.

Conclusion

Humanness is complex: tone, emotion, and style all matter.

More tuning ≠ more human: DPO sometimes reduced authenticity.

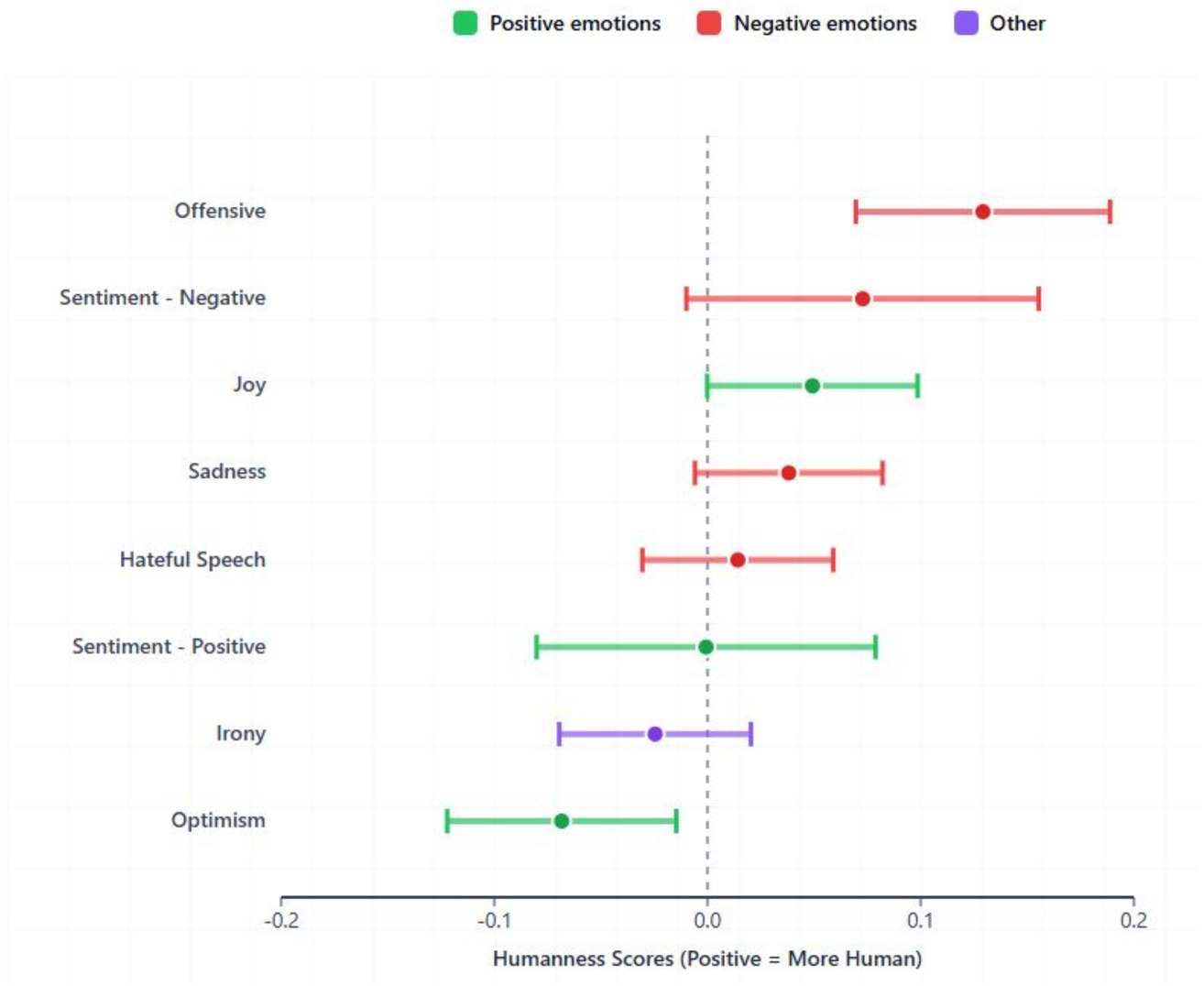
DPO-Subset helped: Smaller, cleaner data improved outputs.

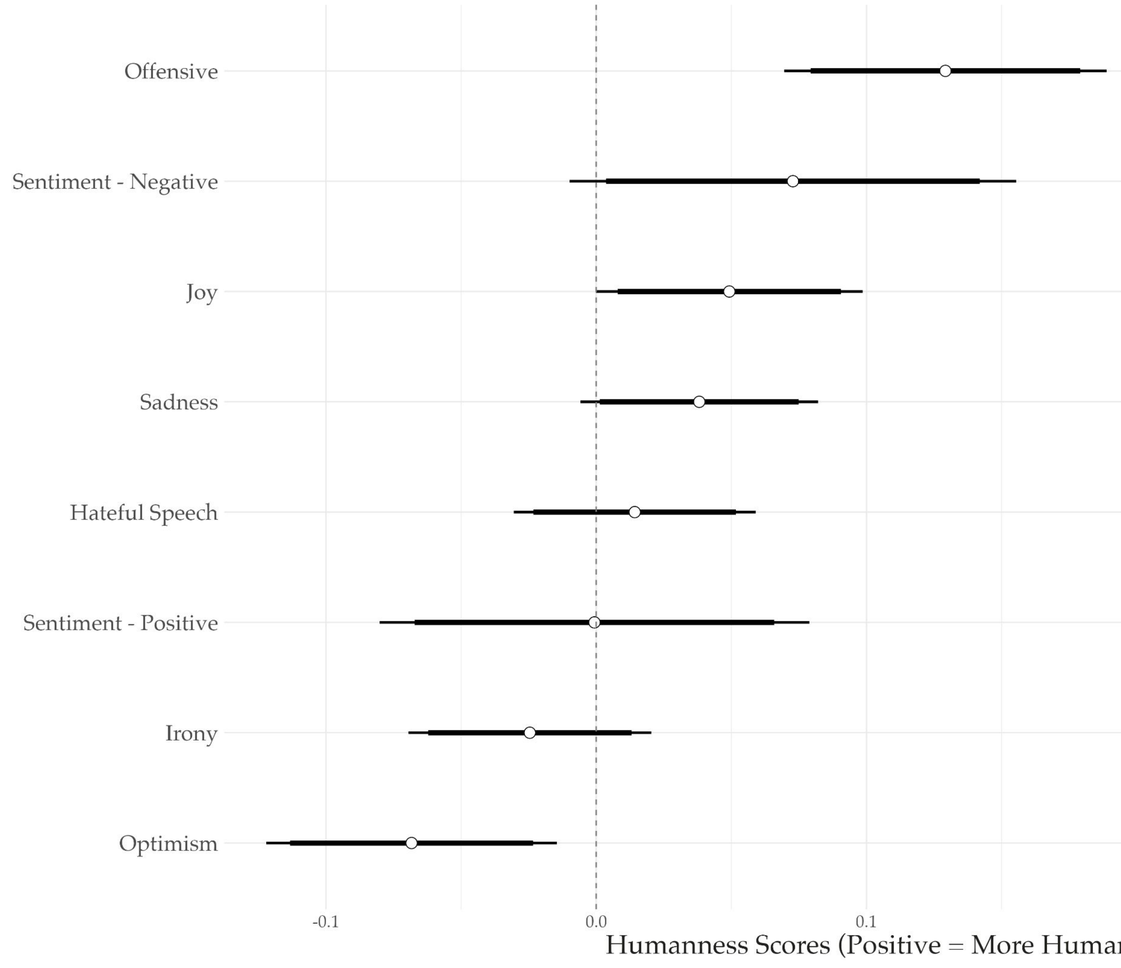
Next steps:

1. Add more platforms (TikTok, TruthSocial)
2. Improve annotation quality (doublel-labeling)
3. Explore RLHF and adaptive sampling to build models

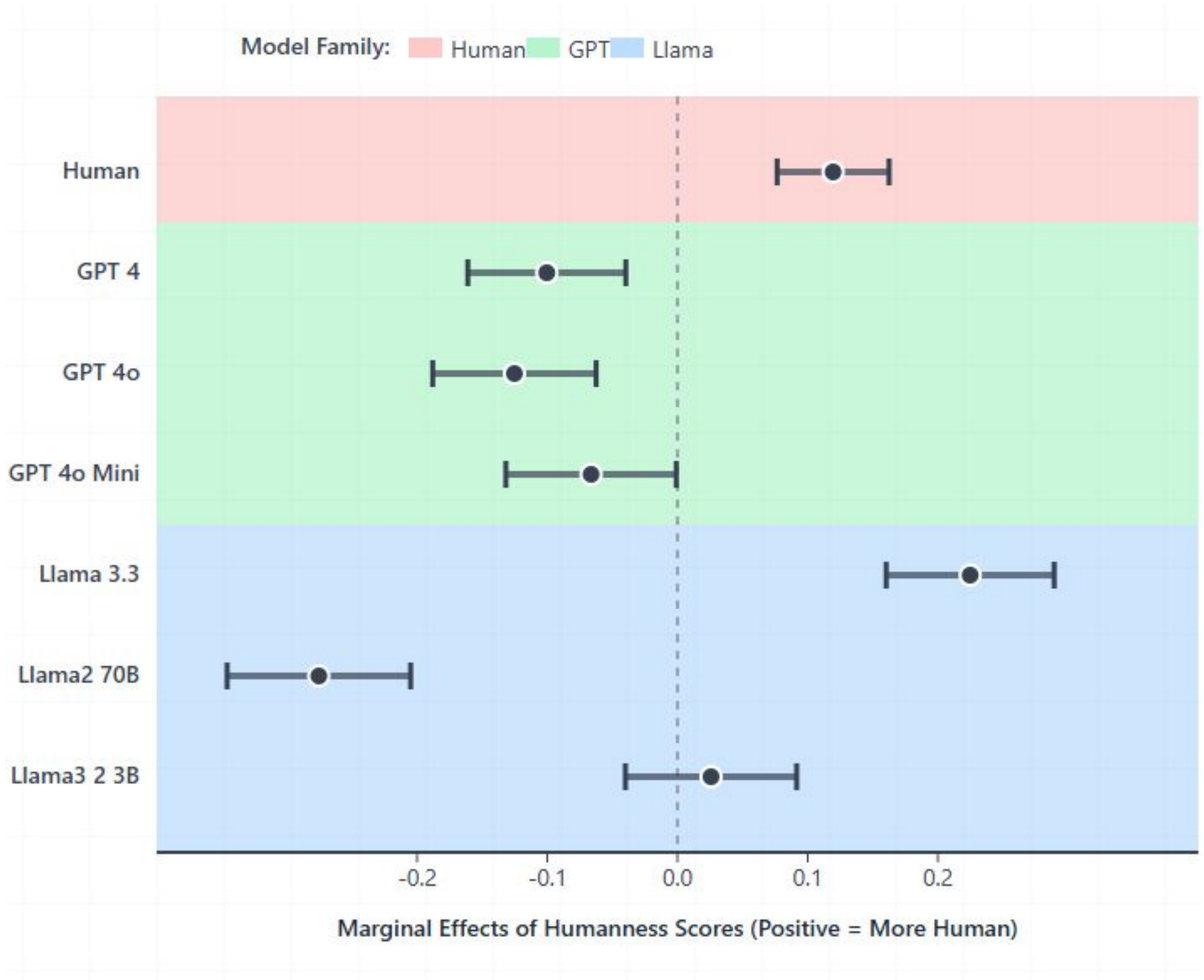
Ethical Considerations

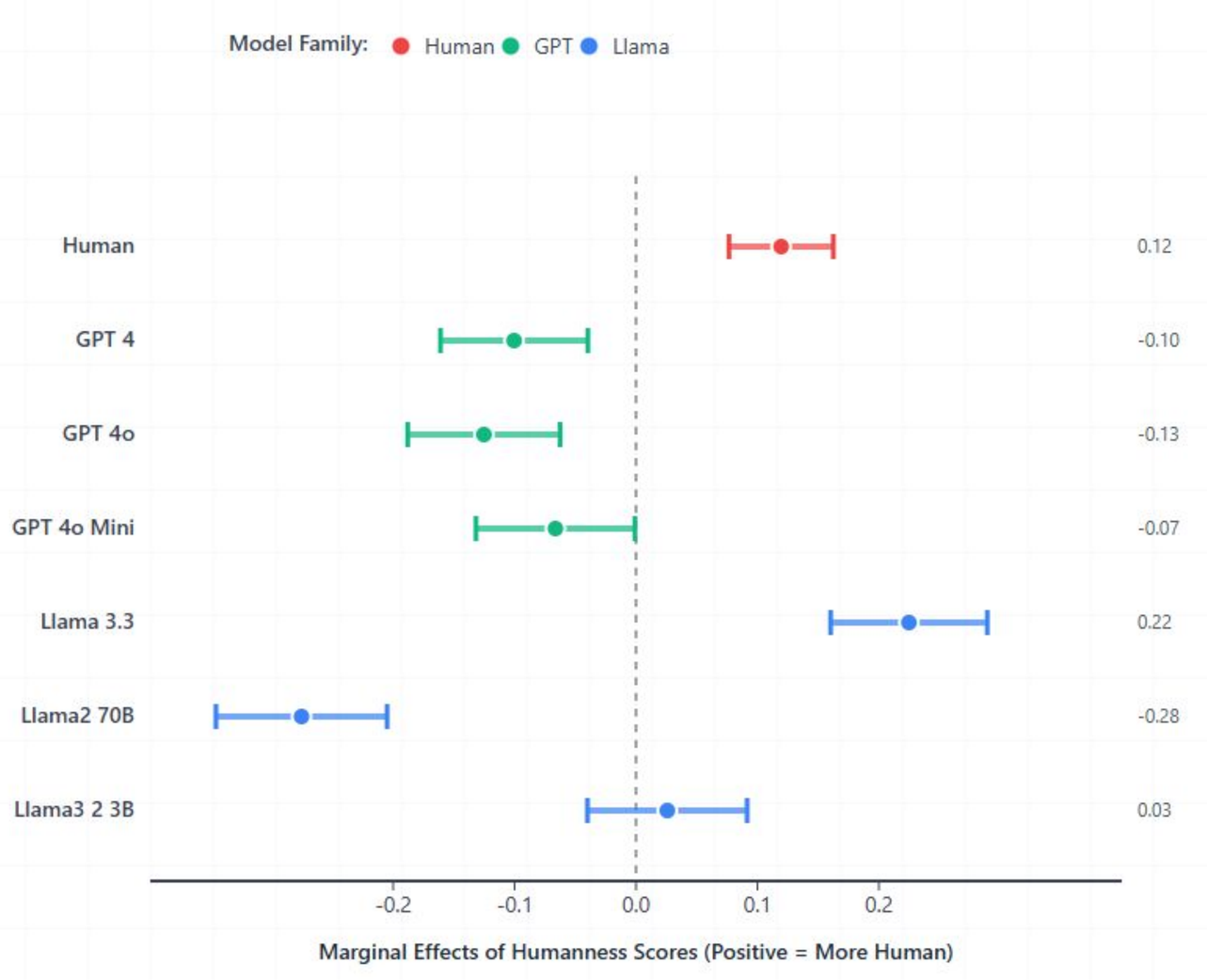
Improving the humanness of AI-generated text may make it harder to distinguish from real content, increasing the risk of misinformation. Our work aims to better understand and control how AI outputs are perceived to support transparency and responsible use.

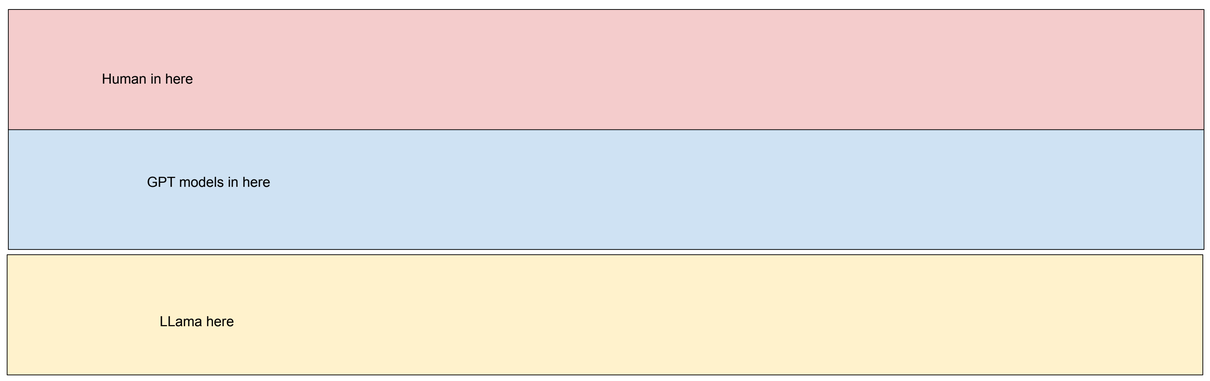

Positive emotions
Negative emotions
Other
Offensive
Sentiment - Negative
Joy
Sadness
Hateful Speech
Sentiment - Positive
Irony
Optimism
-0.2
-0.1
0.0
0.1
0.2
Humanness Scores (Positive = More Human)


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Sentiment - Negative
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Humanness Scores (Positive = More Human

Positive emotions
Negative emotions
Other


Model Family: Human GPT Llama
Human
GPT 4
GPT 4o
GPT 4o Mini
Llama 3.3
Llama2 70B
Llama3 2 3B
-0.2
-0.1
0.0
0.1
0.2
Marginal Effects of Humanness Scores (Positive = More Human)


Model Family: Human GPT Llama
Human
GPT 4
GPT 4o
GPT 4o Mini
Llama 3.3
Llama2 70B
Llama3 2 3B
0.12
-0.10
-0.13
-0.07
0.22
-0.28
0.03
-0.2
-0.1
0.0
0.1
0.2
Marginal Effects of Humanness Scores (Positive = More Human)


Human in here
GPT models in here
LLama here


Pooled Estimates
Human
GPT 4
GPT 4o
GPT 4o Mini